

ENGINEERING MATHEMATICS-I [22MA110]

UNIT – IV

Differential Equations-I: Order and first degree differential equations: Linear differential equations and Bernoulli's equations. Exact differential equations. Applications: Orthogonal trajectories (Cartesian form), Newton's law of cooling, flow of electricity, law of decay and growth.

8+2=10 Hours

1. $\frac{dy}{dx} + y \cot x = \cos x$
2. $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$
3. $2y' \cos x + 4y \sin x = \sin 2x$, given $y=0$ when $x=\pi/3$
4. $(1+y^2)dx = (\tan^{-1} y - x)dy$
5. $dr + (2r \cot \theta + \sin 2\theta)d\theta = 0$

Bernoulli's equation

1. $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$
2. $x \frac{dy}{dx} + y = x^3 y^6$
3. $\frac{dy}{dx} + y \tan x = y^3 \sec x$
4. $\frac{dy}{dx} + \frac{y \log y}{x} = y \left(\frac{\log y}{x} \right)^2$
5. $(x^3 y^2 + xy)dx = dy$
6. $x(x-y)dy + y^2 dx = 0$
7. $\sec^2 y \frac{dy}{dx} + x \tan y = x^3$

Exact Differential equations

1. $(5x^4 + 3x^2 y^2 - 2xy^3)dx + (2x^3 y - 3x^2 y^2 - 5y^4)dy = 0$
2. $(3x^2 + 6xy^2)dx + (6x^2 y + 4y^3)dy = 0$
3. $\left[y \left(1 + \frac{1}{x} \right) + \cos y \right] dx + [x + \log x - x \sin y] dy = 0$

4. $\left(1 + e^{\frac{x}{y}}\right)dx + e^{\frac{x}{y}}\left(1 - \frac{x}{y}\right)dy = 0$
5. $(y^2 e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0$
6. $ye^{xy}dx + (xe^{xy} + 2y)dy = 0$
7. $(y^3 - 3x^2y)dx - (x^3 - 3xy^2)dy = 0$
8. $\frac{dy}{dx} + \frac{(x + 3y - 4)}{(3x + 9y - 2)} = 0$
9. $\frac{2x}{y^3}dx + \frac{y^2 - 3x^2}{y^4}dy = 0$

Orthogonal Trajectories

1. Find the orthogonal trajectories of the family of following curves:
 - i) $y^2 = 4ax$, where a is the parameter
 - ii) $ay^2 = x^3$, where a is the parameter
 - iii) $\frac{x^2}{a^2} + \frac{y^2}{a^2 + \lambda} = 1$, where λ is the parameter
 - iv) $x^2 + y^2 = a^2$, where a is the radius of a circle.

v) Applications of Differential Equations

1. A body originally at 80°C cools down to 60° in 20 minutes, the temperature of air being 40°C . What will be the temperature of the body after 40 minutes from the original?
2. Show that the differential equation for the current I in an electrical circuit containing an inductance L and a resistance R in series and acted on by an electromotive force $E \sin \omega t$ satisfies the equation $L \frac{di}{dt} + Ri = E \sin \omega t$.
3. Uranium disintegrates at a rate proportional to the amount that present at any instant. If M_1 and M_2 grams of uranium are present at times T_1 and T_2 respectively, find the half-life of uranium.